

Backhoe Loader and Excavator Detection in Construction Images

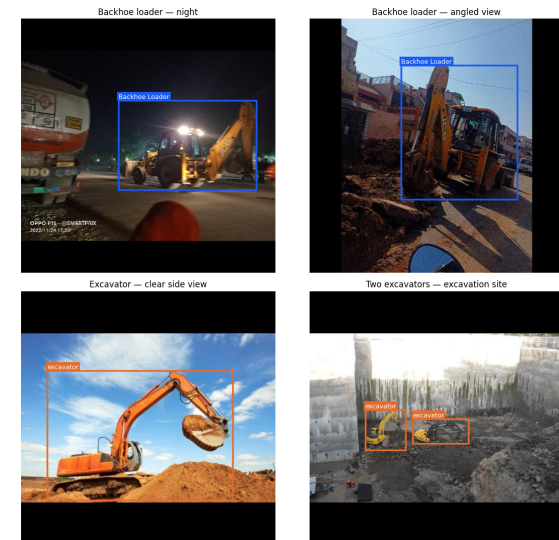
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PROJECT QUESTION

Can a model locate and identify **backhoe loaders** and **excavators** in construction site photos?

A supervisor reviewing equipment costs can compare the machine types detected in dated site photos with hire records when checking the equipment cost sheet. Hire duration and use come from those records.

Evaluation aim: grouped 80/20 split, precision and mAP reporting, plus a provisional recall target of at least 0.70.



Dataset cloning and annotation review

Two target classes: Backhoe Loader and excavator. Images without either class remain negatives.

DataCluster Labs: Backhoe Loaders

Selected source images labeled “excavator” were visually identified as backhoe loaders and relabeled.

Mohamed Sabek: Excavators

Actual excavator photos were cloned from a second source to complete the two-class task.

105

Training

80.2% of 131 images

13

Validation

Held out for selection

13

Testing

Untouched final check

11

Negatives

Across all three splits

Frozen version 3: about 80% train and 20% held out (validation + test). Related views were grouped before export. Auto-orient, 640 × 640 black-edge fit and exclusion-tag filtering were applied; export augmentation was off.

Photo sources: DataCluster Labs and Mohamed Sabek on Roboflow Universe; CC BY 4.0 as listed by each source.

From Roboflow versions to the trained detector

Related views were grouped before the frozen version 3 export.

V1 review

Labels and split inspected

Group views

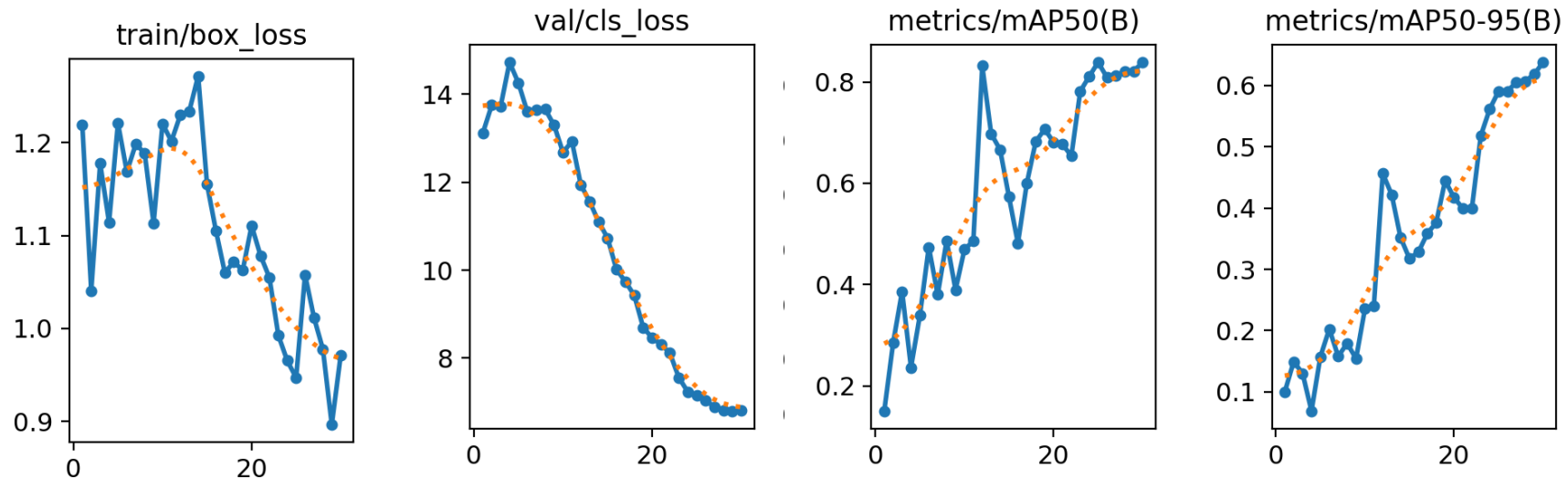
Related photos kept together

Freeze V3

131 images and SHA256 ZIP

Train + test

YOLO26n, 30 epochs; 13 test photos



Selected YOLO curves: training box loss and validation class loss fell overall; validation mAP50 and mAP50-95 rose. The full results.png is in the report and repository.

Test results on 13 held-out images

Twelve labeled objects; the 0.70 provisional recall aim was exceeded in this small test.

0.812

Precision

Predicted box correctness

0.904

Recall

Provisional aim: 0.70

0.938

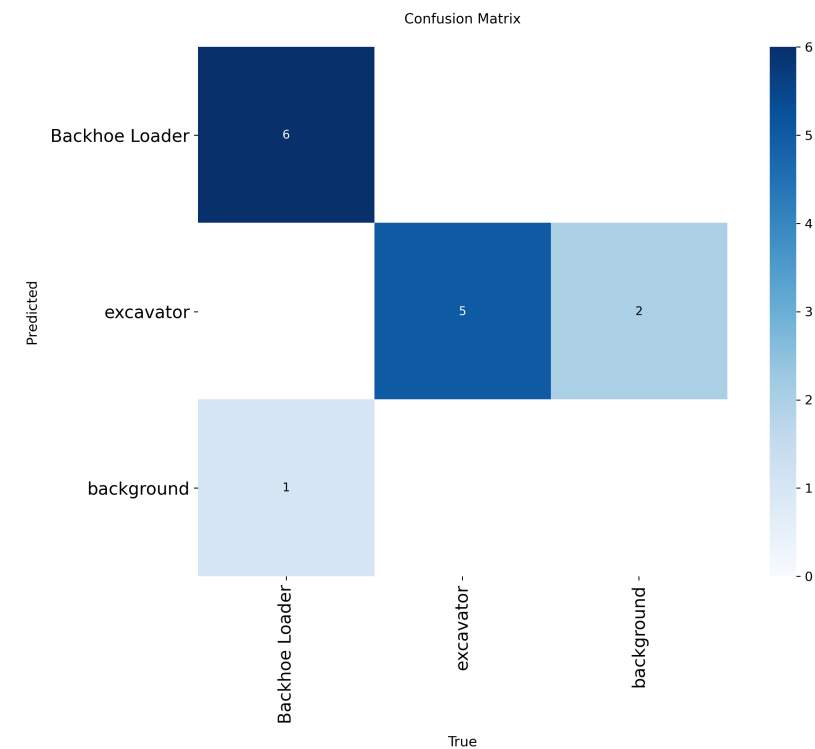
mAP50

IoU threshold 0.50

0.677

mAP50-95

Across stricter overlaps



With only 12 labeled test objects, each miss or extra box can change the summary scores noticeably.

“Background” records unmatched labels or predictions during evaluation. It is not a third class.

Correct detections and error cases

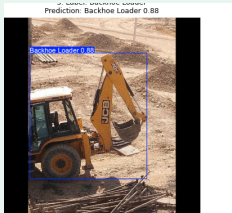
Selected predictions at confidence 0.25; the repository documents three FP detections and three FN examples.

WHAT WORKED



Rear-view backhoe

Backhoe Loader 0.85



Partial backhoe

Backhoe Loader 0.88



Site excavator

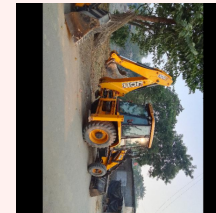
excavator 0.86

WHAT FAILED



Two False Positives

Negative tractor: Backhoe Loader 0.37; excavator 0.35.



False negative

Sideways backhoe loader: no detection.



Duplicate False Positive

One excavator: two overlapping boxes.

Investigation of a missed backhoe loader

The labeled target received no box. Auto-orient was enabled, but this exported photo remained sideways.



As exported: no detection at confidence 0.25. **Rotated upright:** Backhoe Loader 0.81 with the same checkpoint. Check image orientation during data cleaning and add realistic tilted views. This diagnostic does not change the original test score.

Workable Detection Prototype

The model detects equipment presence in individual images, with known limits.

Three data improvements tied to the observed errors:

1 Correct orientation and broaden views

Sideways and transported targets were missed. Verify upright files; add labeled tilted, partial and transport views.

2 Strengthen hard negatives

A tractor received two target boxes. Add unboxed tractors and similar loaders in comparable scenes.

3 Review boxes and diversify scenes

An excavator drew a duplicate box. Audit box consistency and expand the small independent test set.

USE WITH HUMAN REVIEW

A supervisor can compare per-photo detections with dated hire records. These images alone do not establish the hours hired or used.

HANDOVER

The public release holds the frozen dataset and best.pt; the repository has notebooks, evidence and source attribution.

Conclusion: The model can help screen individual site photos for these two machine types. Human review and wider testing remain essential before using detections in a cost review.